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# INTERNSHIP PROJECT REPORT

## Sentiment Analysis Using Natural Language Processing

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| Submitted By        | Ayush Kashyap                  |
| Internship Role     | Artificial Intelligence Intern |
| Organization        | Codec Technologies Pvt. Ltd.   |
| Internship Duration | 01 July 2026 – 30 July 2026    |

Submitted in partial fulfillment of the internship requirements

Academic Year 2026

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## 1. Executive Summary

This report presents the work and learning outcomes of my one-month Artificial Intelligence internship at Codec Technologies Pvt. Ltd. The internship provided structured exposure to Artificial Intelligence, Python programming, data preprocessing, machine learning, deep learning, natural language processing, and project development.

For the capstone project, I selected Sentiment Analysis Using Natural Language Processing (NLP). The project focuses on automatically identifying the sentiment expressed in textual data and classifying it into meaningful categories such as positive, negative, or neutral.

## 2. Introduction

Artificial Intelligence enables computer systems to perform tasks that normally require human intelligence, including learning from data, recognizing patterns, making predictions, and understanding language. Natural Language Processing is a major branch of AI that helps computers process and interpret human language.

Sentiment analysis is a practical NLP application used to understand opinions expressed in customer reviews, social-media posts, feedback forms, and other textual sources. This project demonstrates a complete basic workflow from text preprocessing to feature extraction, model training, prediction, and evaluation.

## 3. Internship Overview

- Designation: Artificial Intelligence Intern
- Organization: Codec Technologies Pvt. Ltd.
- Duration: 01/07/2026 to 30/07/2026
- Mode: Pan India, Hybrid
- Training coverage: AI fundamentals, Python, data handling, machine learning, deep learning, NLP, reinforcement learning, and capstone development.

## 4. Problem Statement

Large volumes of text are generated every day through online reviews, surveys, comments, and social platforms. Manually reading and categorizing this information is time-consuming and inconsistent. The problem addressed in this project is to build an automated system that can process textual input and predict the sentiment expressed by the user.

## 5. Project Objectives

- Understand the basic workflow of an NLP-based machine learning project.
- Clean and preprocess raw text data.
- Convert text into numerical features using TF-IDF.
- Train a supervised machine learning classifier.
- Predict positive, negative, or neutral sentiment for new text.
- Evaluate model performance using standard classification metrics.

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## 6. Tools and Technologies

- Python – core programming language.
- NumPy and Pandas – data manipulation and analysis.
- NLTK – basic natural language processing support.
- Scikit-learn – feature extraction, model training, and evaluation.
- Matplotlib / Seaborn – exploratory visualization where required.
- Jupyter Notebook or similar Python environment – implementation and experimentation.

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## 7. Methodology

The project follows a standard machine learning pipeline. Each stage prepares the data for the next stage and helps make the final prediction process systematic and reproducible.

| Step | Activity                                                               |
|------|------------------------------------------------------------------------|
| 1    | Collect or prepare labelled text data                                  |
| 2    | Clean text by removing noise and unnecessary characters                |
| 3    | Normalize text using lowercasing, tokenization, and stop-word handling |
| 4    | Convert text into numerical features using TF-IDF                      |
| 5    | Split data into training and testing sets                              |
| 6    | Train a classification model such as Logistic Regression               |
| 7    | Evaluate using accuracy, precision, recall, and F1-score               |
| 8    | Use the trained model to predict sentiment for new text                |

## 8. Data Preprocessing

Raw text is often noisy and cannot be directly used by a machine learning model. Preprocessing improves consistency and reduces irrelevant variation. Typical operations include converting text to lowercase, removing punctuation and special characters, tokenizing sentences into words, handling common stop words, and preparing clean text for feature extraction.

## 9. Feature Extraction Using TF-IDF

Machine learning models require numerical input. TF-IDF (Term Frequency–Inverse Document Frequency) converts text into a numerical representation based on the importance of words in a document relative to the overall dataset. Frequently occurring but less informative words receive lower importance, while more distinctive terms can receive greater weight.

## 10. Model Development

A supervised classification approach is used. After transforming the cleaned text into TF-IDF features, the dataset is divided into training and testing sets. A classification algorithm such as Logistic Regression can learn relationships between textual features and sentiment labels. Once trained, the model can estimate the sentiment class of previously unseen text.

## 11. Model Evaluation

- Accuracy: overall proportion of correct predictions.
- Precision: proportion of predicted positive instances that are correct.
- Recall: ability to identify relevant instances.
- F1-score: balanced measure combining precision and recall.

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- Confusion matrix: detailed view of correct and incorrect classifications.

## 12. Sample Project Workflow

Input Text → Text Cleaning → Tokenization / Normalization → TF-IDF Vectorization → Trained Classification Model → Predicted Sentiment

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## 13. Learning Outcomes

- Developed a stronger understanding of Artificial Intelligence and Machine Learning fundamentals.
- Improved practical Python programming skills for data-oriented tasks.
- Learned data cleaning, preprocessing, and visualization concepts.
- Understood supervised learning and model evaluation techniques.
- Gained introductory exposure to neural networks, TensorFlow, Keras, and CNN concepts.
- Developed practical understanding of NLP concepts and sentiment analysis.
- Learned how to structure, document, and present an end-to-end AI project.

## 14. Challenges and Solutions

- Noisy text data: addressed through preprocessing and normalization.
- Converting language into model input: handled through TF-IDF feature extraction.
- Choosing evaluation measures: addressed by using multiple classification metrics rather than relying only on accuracy.
- Understanding the complete workflow: improved by following a step-by-step pipeline from data preparation to prediction.

## 15. Applications

- Customer review analysis
- Product and service feedback monitoring
- Social media opinion analysis
- Brand reputation tracking
- Survey response analysis
- Basic support and feedback prioritization

## 16. Future Scope

- Use larger and more diverse datasets.
- Experiment with advanced deep learning architectures.
- Use transformer-based language models for improved contextual understanding.
- Build a web interface where users can enter text and receive sentiment predictions.
- Extend the system to multilingual sentiment analysis.

## 17. Conclusion

The Artificial Intelligence internship provided a structured learning experience covering important concepts from AI foundations to advanced applications. The Sentiment Analysis project helped connect these concepts into a practical workflow. Through this project, I learned how text can be prepared, transformed

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into features, used to train a machine learning model, and evaluated to produce meaningful predictions.

Overall, the internship strengthened my understanding of Python, data handling, machine learning, NLP, and AI project development. The experience has provided a foundation for further work on more advanced artificial intelligence and data science projects.

## 18. References

- Python Documentation
- Scikit-learn Documentation
- NLTK Documentation
- NumPy and Pandas Documentation
- Internship learning material and training modules

## 19. Declaration

I hereby declare that this project report titled “Sentiment Analysis Using Natural Language Processing” is prepared as a summary of my internship learning and project work. The report has been compiled for academic submission and presentation purposes.

**Submitted by:** Ayush Kashyap

**Internship Role:** Artificial Intelligence Intern